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TASK 01 – UK SPONSORSHIP VISA FOR EMPLOYERS

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1. Introduction

The UK has a sponsorship visa regime that helps international migrants move into the country for reasons, with the most common ones being university education and job opportunities. In respect to the sponsorship visa regime, the licensed institutions, organizations, and employers are responsible for sponsoring individuals, with the sponsorship resulting in the issuing of sponsorship documents that are essential for potential immigrants to acquire visas. The sponsorship document in respect to university education is called the CAS, while for job opportunities it is called CoS.

The past few years have caused a dramatic shift in the way that people migrate within the UK. This is not limited to certain factors, such as the changes in immigration law introduced after Brexit, but also includes overall global economic conditions and labor shortages, together with the aftermath of the COVID-19 pandemic. The effects on sponsorship of education and employment have meant that the analysis of sponsorship data is a pivotal point for consideration.

The ambition of this project is to use a blend of SQL Server for storing, cleaning, and preparing historical sponsorship visa data, coupling it with Microsoft Power BI for modeling, visualizing, and deriving insights from that data. This particular project aims to make sense of long-run trends, quarterly behavior, geographically/nationally distinct activity, as well as comparisons based on sponsorship types that are work versus university-based.

2. Data Source & Extraction

The analysis draws on official UK sponsorship visa datasets. The datasets were obtained from Data.gov.uk, the central repository for UK government open data. The data consists of four core tables, each capturing different angles of study and work sponsorships:

Study Sponsorship Tables

CAS_D01: Annual and quarterly profile of applications for study sponsorship by type of application, institution type group and type of institution

CAS_D02: This has similar coverage to the study sponsorships but adds the geographic region and nationality to the year/quarter/application type/institution type group dimensions.

Work Sponsorship Tables

CoS_D01: Number of work sponsorship applications by year and quarter, by application type, leave category and industry

CoS_D02: Work sponsorships tracked by year and quarter and application type, with geographic region and nationality included.

3. SQL Server Implementation

A separate SQL Server database was created for storing and managing sponsorship data and it was named Uk_Immigration. Using the Import Flat File in SQL Server, the CSV files were imported into SQL Server, with each data set linked to its respective table.

3.1 Data Cleaning & Transformation

One of the major issues that were found to affect data quality while importing the data was that the applications column in the four tables was represented by the data type of NVARCHAR, since it contains comma-separated values like "1,234." This is a major issue since numerical analysis is carried out using integer values. We removed the comma and then changed the data type to int.

```
-- Removing commas and turning updationg application column data type in all tables

UPDATE CAS_D01
SET Applications = REPLACE(Applications, ',', '');
ALTER TABLE CAS_D01
ALTER COLUMN Applications INT;

UPDATE CAS_D02
SET Applications = REPLACE(Applications, ',', '');
ALTER TABLE CAS_D02
ALTER COLUMN Applications INT;

UPDATE CoS_D01
SET Applications = REPLACE(Applications, ',', '');
ALTER TABLE CoS_D01
ALTER COLUMN Applications INT;

UPDATE CoS_D02
SET Applications = REPLACE(Applications, ',', '');
ALTER TABLE CoS_D02
ALTER COLUMN Applications INT;

-- remove blanks in application column

UPDATE CAS_D01 SET Applications = NULL WHERE Applications = '';
UPDATE CAS_D02 SET Applications = NULL WHERE Applications = '';
UPDATE CoS_D01 SET Applications = NULL WHERE Applications = '';
UPDATE CoS_D02 SET Applications = NULL WHERE Applications = '';
```

To achieve proper chronological sorting and mathematical calculations through time, the numeric ordering of quarters became relevant. This involved using a calculated quarter number to signify the order of the quarters through the years instead of using text values for the quarters.

```sql
-- Clean Quarter column (extract quarter number)

ALTER TABLE CAS_D01
ADD QuarterNumber AS CAST(RIGHT(Quarter, 1) AS INT) PERSISTED;

ALTER TABLE CAS_D02
ADD QuarterNumber AS CAST(RIGHT(Quarter, 1) AS INT) PERSISTED;

ALTER TABLE CoS_D01
ADD QuarterNumber AS CAST(RIGHT(Quarter, 1) AS INT) PERSISTED;

ALTER TABLE CoS_D02
ADD QuarterNumber AS CAST(RIGHT(Quarter, 1) AS INT) PERSISTED;


-- Create QuarterSort column

ALTER TABLE CAS_D01
ADD QuarterSort AS (Year * 10 + CAST(RIGHT(Quarter, 1) AS INT)) PERSISTED;

ALTER TABLE CAS_D02
ADD QuarterSort AS (Year * 10 + CAST(RIGHT(Quarter, 1) AS INT)) PERSISTED;

ALTER TABLE CoS_D01
ADD QuarterSort AS (Year * 10 + CAST(RIGHT(Quarter, 1) AS INT)) PERSISTED;

ALTER TABLE CoS_D02
ADD QuarterSort AS (Year * 10 + CAST(RIGHT(Quarter, 1) AS INT)) PERSISTED;
```

After that all the data columns were standardized to get the same format throughout all the tables.

Then to make it easier to report, a unified SQL view was developed for study sponsorships and work sponsorships. This is achieved by using the UNION ALL command to combine data from the respective D01 or D02 table. It should be noted that the SQL view is not designed to remove null values but to provide a standardized fact structure for easier interpretation by Power BI. Notably, the nulls were retained in the consolidated data views since they represent real differences in the data structure between datasets, as opposed to missing or incorrect data.

```sql
CREATE OR ALTER VIEW Study_Sponsorship AS
SELECT
     Year,
     Quarter,
     QuarterNumber,
     QuarterSort,
     Type_of_application,
     Institution_type_group,
     Institution_type,
     NULL AS Geographical_region,
     NULL AS Nationality,
     Applications
FROM CAS_D01

UNION ALL

SELECT
     Year,
     Quarter,
     QuarterNumber,
     QuarterSort,
     Type_of_application,
     Institution_type_group,
     NULL AS Institution_type,
     Geographical_region,
     Nationality,
     Applications
FROM CAS_D02;
```

4. Power BI data connection and Transformation

The tables were imported directly to power bi by connecting them and we chose the 2 unified views tables for our analysis

Only light transformations were performed with Power BI since majority of data cleaning was carried out in SQL Server. The major transformations with Power BI involved,

Verifying data types

Renaming columns

Creating calculated measures using DAX –

Total Applications =

[Total Study Applications] + [Total Work Applications]

Total Study Applications =

SUM (Fact_Study[Applications])

Total Work Applications =

SUM (Fact_Work[Applications])

Study to Work Ratio =

DIVIDE ([Total Study Applications], [Total Work Applications])

Study % Share =

DIVIDE ([Total Study Applications], [Total Applications])

Work % Share =

DIVIDE ([Total Work Applications], [Total Applications])

These are some of the DAX measures that were calculated for more precise and important visualizations.

Notably, the null transformations were not removed. This was done differently, where nulls were managed through graph-level filters for precise analysis.

5. Data Model & Relationships

The Power BI Data Model consisted of a star-schema structure that included the following:

- Fact tables: Unified study sponsorship data, Unified work sponsorship data

- Dimension tables include Date, Nationality, Geographical Region, Institution Type, Industry, and Leave Category.

We established relationships between the fact table and the dimension tables using one-to-many cardinality where possible. In the dimension tables, there were unique values for each attribute, whereas the fact tables consisted of multiple records related to the attributes. Cross-filtering remained mostly one-way to prevent ambiguity. Many-to-many associations were avoided since they tend to introduce complexity that could influence the results of aggregations. Some of the attributes were omitted from the dimension tables either on account of too many nulls or duplications. Instead of imposing artificial connections, such attributes were evaluated in fact-level graphics.

6. Dashboard Visualization Analysis

6.1 Overview

- The total applications card shows combined total volume of Study and Work sponsorship applications reached 16 million. This large volume underscores the UK's significance as a destination for international students and workers.
- Study to work ratio card represents that for every 1.74 Study applications, there is 1 Work application. This indicates that the Study stream (CAS) is significantly larger than the Work stream (CoS), which is typical for global migration trends.
- Latest quarter applications card shows a volume of 328,000 applications in the latest quarter shows strong, sustained demand for sponsored visas.
- Total study and work applications by year line chart shows both Study and Work applications show a dramatic and steep upward trend, particularly starting around 2020. Study applications consistently outpaced Work applications. The line for Total Study Applications is significantly higher over the years. Clear increase post-2020 suggests a strong, positive response to recent UK immigration policy changes designed to attract students and skilled workers.
- Total applications by geographical regions shows south Asia leads the applications with around 2 million applications followed by east Asia and sub Saharan Africa.
- Study and Work QoQ growth by year represents both Study QoQ Growth and Work QoQ Growth show high volatility across the years, but with overall high growth rates visible by 2020. Growth rate frequently surpasses 100% and reaches nearly 400% in some periods for both categories, indicating very rapid expansion. The high growth percentages (rather than volume) confirm that recent years have seen an exponential increase in demand, not just a gradual rise.

6.2 Study Sponsorship

- Total study applications has 10M applications confirming the Study stream is the largest component of the overall 16M applications.
- Study % share card shows study applications account for 64% of the total combined application pool.
- Study QoQ growth card show the latest Quarter-over-Quarter growth rate is exceptionally high at 1692.82%, emphasizing the explosive demand in the most recent periods.
- Top Study Applications by Nationality bar chart represent top 3 Nationalities where China leads significantly with 1,614,458 applications. India follows with 782,071 applications. Nigeria ranks third with 265,997 applications. The Top 3 countries (China, India, Nigeria) account for the vast majority of sponsored study visa demand, a key finding for the "Top 10 Nationalities" requirement.
- Total Study Applications by Institution type group bar chart shows the 'Higher Education' group (Universities) is overwhelmingly dominant. This group accounts for the bulk of the 10M total study applications, confirming the primary attraction of the UK's university sector. The 'All other institutions' group shows a smaller, but still significant, volume, representing demand across colleges, schools, and private providers.
- Total Study Applications by Institution type shows 'Non Russell Group' and 'Russell Group' institutions have the highest volumes, consistent with the dominance of the university sector. Categories like 'Tertiary, Further Edu' show the next highest volumes, confirming that application volume is spread across different educational levels.

6.3 Work Sponsorship

- Top work applications card confirms the Work stream's volume, making up the remaining 6M of the overall 16M applications.
- Work % share card shows work applications account for 36% of the total combined application pool.
- Work QoQ growth shows the latest Quarter-over-Quarter growth for the Work stream is extremely high at 2728.96%. This suggests that work migration policies have seen a massive, recent surge in utilization.
- Total Work Applications by Nationality shows us the top 3 Nationalities. India leads the work sponsorship demand with 872,585 applications. United States ranks second with 394,420 applications. Nigeria ranks third with 170,635 applications. Unlike the Study stream (led by China), the Work stream is led by India, followed by applicants from the United States, indicating a strong demand for skilled labor from North America.
- Total Work Applications by Industry represent leading Industries - 'Human Health and Social Work' and 'Arts, Entertainment, and Recreation' show the highest volumes of work applications. Other Major Sectors - 'Information and Communication' and 'Professional, Scientific, and Technical Activities' also represent major industry users of the sponsorship system, confirming the demand for highly skilled labor.
- Total Work Applications by Category of Leave show dominance of Skilled Worker: 'Worker (Previous Tier 2)' visa category (the Skilled Worker route) is the overwhelming category for work applications, with volumes significantly higher than 2M. Temporary vs. Skilled: The 'Temporary' worker category has a much lower volume, indicating that the

vast majority of sponsored work applications are for long-term, skilled employment, rather than short-term placements.

7. Key Findings & Insights

7.1 Study sponsorships

One of the few outstanding results is the consistent prevalence of applications for study sponsorship over work sponsorships during the whole period under review. According to the Overview dashboard, study-related applications account for approximately two-thirds of all sponsorship activities, a trend reemphasized by the Study % Share KPI and a Study-to-Work Ratio standing at approximately 1.7.

This points to international education as the main driver of UK-sponsored migration. The consistently higher total study applications suggest this dominance is structural, not fleeting. UK universities continue to attract large numbers of international students, placing education at the heart of the country's migration and economic strategy.

7.2 QoQ Dynamics

Quarter-on-quarter growth analysis shows several notable volatilities, particularly in study sponsorships. In some periods, the QoQ growth is remarkably high and exceeds 1000% or more. Although these look odd at first sight, they mathematically make sense and are due to very low baselines in the previous quarter.

Work sponsorships also reflect QoQ volatility, with movements generally less pronounced than those in study sponsorships. This reflects the nature of work visas in response to short-term labor market conditions, employer demand, and broader economic cycles.

Overall, QoQ growth metrics may be useful for short-term insight but should be read together with absolute values and long-term trends to avoid misinterpretation.

7.3 Nationality Concentration

Nationality-based analysis across both study and work dashboards reveals a strong volume concentration in sponsorship applications from a small set of countries. In study sponsorships, the top contributors usually come from China, India, Nigeria, Pakistan, and the United States. For work sponsorships, India is usually the leading source, with just a few other countries contributing volumes that are much smaller.

This level of concentration creates vulnerabilities to dependence. The overall volume of sponsorship in the UK may be disproportionately affected by changes in geopolitical relations, source-country policies, or shifts in global student mobility. The results indicate that while the UK has a strong supply from several key countries, source diversification would strengthen resilience over the longer term.

7.4 Institutional and Sectoral Dependence

The sponsorship issuance is strongly driven by higher education institutions, according to the study sponsorship dashboard. Non-Russell Group and Russell Group universities make up the lion's share of the applications for studies, proof of the UK's high standing internationally in higher education.

The opposite of this is that the work sponsorships are focused in industries. Healthcare, information technology, professional services, and associated others also make up the lion's share of the work sponsorship activity. The large prevalence of healthcare-related sponsorships probably reflects the continuing skill shortfall, aging demography, and structural hiatus of the UK workforce.

Category-of-leave analysis further shows a small set of sponsored work routes driving the biggest application volumes, suggesting low diversification in work-based migration pathways.

7.5 Regional Patterns of Sponsorship Demand

The regional breakdown indicates that applications for sponsorship to both study and work originate overwhelmingly from Asia, particularly from South and East Asia. A considerable share also emanates from Africa, while the Americas, Europe (apart from the UK), and the Middle East contribute a relatively smaller number. The distribution of these patterns across the dashboards, and in a manner that was consistent with the UK's strong ties to some global regions, underscores regional imbalances that might shape future migration strategy and international engagement.

8. Conclusion

This report demonstrates a step-by-step walkthrough of an entire data analytics process, using SQL Server and Power BI for analysis of support sponsorship visas issued in the UK. SQL Server is a robust foundation for data storage, processing, and merging, ensuring data integrity is maintained consistently from one stage of analysis to the next.

Finally, the implementation of Power BI is used to create a robust data model with appropriate relationships. This creates a multi-page dashboard with both broad executive summaries and

detailed analysis. These dashboards accommodate long-term trends, recent changes, nationality clusters, industry dependency, and sponsorship application geographic diffusion.

Analysis of the data reveals that study sponsorships remain the crucial sponsorship type driving UK-sponsored migration, while work sponsorships remain important but more volatile, mirroring labor market trends. Findings suggest areas of both strength and weakness in the current sponsorship scheme, especially regarding sponsorship by source countries and source industries.